CUSTOMER CHURN ANALYSIS REPORT

Telecom Industry · 7,043 Customer Records

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Executive Summary

This report presents a comprehensive analysis of customer churn for a telecom company with 7,043 active and former customers. The analysis uncovers critical patterns behind customer attrition, quantifies the financial impact, and provides actionable recommendations to reduce churn.

Key Findings at a Glance

7,043	1,869	26.54%	$16.06M	$64.76	$139K
Total Customers	Churned Customers	Churn Rate	Total Revenue	Avg Monthly Charges	Monthly Revenue Lost

The overall churn rate of 26.54% is significantly above the telecom industry benchmark of 15–20%, indicating an urgent need for targeted retention strategies. Month-to-month contract customers, Fiber optic internet users, and Electronic check payment users are the highest-risk segments. Early-tenure customers (0–12 months) are churning at a staggering 47.4%, pointing to a critical gap in the onboarding experience.

1. Business Problem

Customer churn — when a customer stops using a company's services — is one of the most critical challenges in the telecom industry. Acquiring a new customer costs 5–7 times more than retaining an existing one, making churn reduction a top financial priority.

1.1 Problem Statement

The company is experiencing a churn rate of 26.54%, resulting in a monthly revenue loss of $139,130 and a cumulative total revenue loss of $2.86 million. Left unaddressed, this trajectory threatens long-term profitability and customer lifetime value.

1.2 Analysis Objectives

- Measure and benchmark the overall churn rate
- Identify the customer segments and attributes most strongly correlated with churn
- Quantify the financial impact of churn across contracts and services
- Detect early warning signals for at risk customers
- Provide data-driven recommendations to reduce churn by at least 5–8%

2. Data Analysis & Key Findings

2.1 Overall Churn Rate

Out of 7,043 total customers, 1,869 have churned — a churn rate of 26.54%. This is notably higher than the industry average of 15–20%, confirming churn as a critical business issue requiring immediate attention.

Metric	Value	Benchmark
Total Customers	7,043	—
Churned	1,869	—
Retained	5,174	—
Churn Rate	26.54%	15–20% (industry avg)
Monthly Revenue Lost	$139,131	—
Total Revenue Lost	$2,862,927	—

2.2 Churn by Contract Type

Contract type is the single strongest predictor of churn. Month-to-month customers churn at 42.71% — 15x higher than two-year contract customers. This reveals a fundamental retention gap in short-term subscription management.

Contract Type	Total Customers	Churned	Churn Rate
Month-to-month	3,875	1,655	42.71% ▲ HIGH
One year	1,473	166	11.27% ◆ MEDIUM
Two year	1,695	48	2.83% ▼ LOW

2.3 Churn by Internet Service

Fiber optic customers exhibit a disproportionately high churn rate of 41.89% despite likely paying premium prices. This suggests a service quality or value perception problem specific to fiber subscribers.

Internet Service	Total Customers	Churned	Churn Rate
Fiber optic	3,096	1,297	41.89% ▲ HIGH
DSL	2,421	459	18.96% ◆ MEDIUM
No internet	1,526	113	7.40% ▼ LOW

2.4 Churn by Payment Method

Electronic check users churn at 45.29% — nearly 3x the rate of automatic payment users. Customers on autopay (credit card or bank transfer) are significantly more likely to remain with the company.

Payment Method	Total Customers	Churned	Churn Rate
Electronic check	2,365	1,071	45.29% ▲
Mailed check	1,612	308	19.11%
Bank transfer (auto)	1,544	258	16.71%
Credit card (auto)	1,522	232	15.24% ▼

2.5 Churn by Tenure — Early vs. Loyal Customers

New customers are leaving at an alarming rate. Nearly half of all customers in their first year (47.44%) churn, while only 9.51% of customers who have been with the company for 49+ months churn. This pattern is a classic signal of a poor onboarding or early-experience problem.

Tenure Bucket	Total Customers	Churned	Churn Rate
0–12 months	2,186	1,037	47.44% ▲ Critical
13–24 months	1,024	294	28.71%
25–48 months	1,594	325	20.39%
49+ months	2,239	213	9.51% ▼ Healthy

2.6 Impact of Add-On Services on Churn

Customers subscribed to both OnlineSecurity and TechSupport have a churn rate of only 9.01% — compared to 48.96% for customers with neither add-on. This 5x difference represents a powerful lever for both retention and upselling.

Online Security	Tech Support	Total	Churn Rate
No	No	2,553	48.96% ▲ Highest risk
No	Yes	945	22.33%
Yes	No	920	21.30%
Yes	Yes	1,099	9.01% ▼ Lowest risk

2.7 Senior Citizen Churn Analysis

Senior citizens (SeniorCitizen = 1) represent only 16.2% of the customer base, yet churn at 41.68% — nearly double the rate of non-senior customers at 23.61%. This demographic requires targeted support and simplified service offerings.

Segment	Total Customers	Churned	Churn Rate
Non-Senior (0)	5,901	1,393	23.61%
Senior Citizen (1)	1,142	476	41.68% ▲

3. Power BI Dashboard

The interactive Power BI dashboard below was built directly from this dataset. It features 5 KPI cards, 4 interactive charts, and 4 slicers (Payment Method, Senior Citizen, Online Security, Contract Type) allowing stakeholders to filter the entire view dynamically.

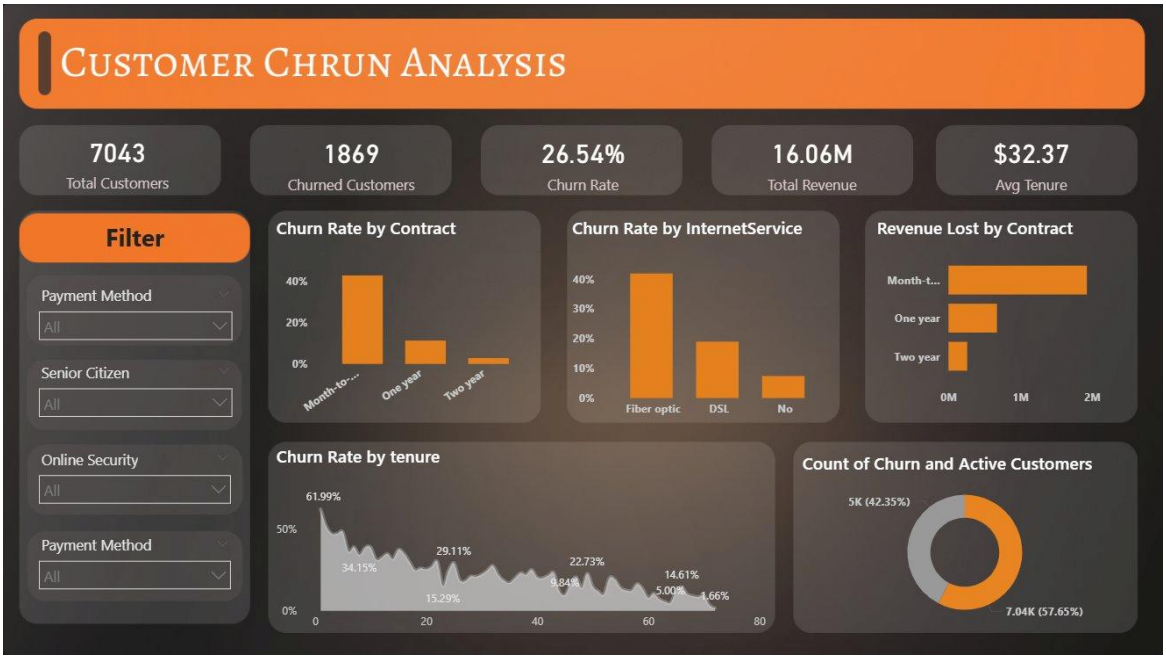

Figure 1: Customer Churn Analysis — Power BI Dashboard

Dashboard highlights:

- Top KPI cards: Total Customers, Churned Customers, Churn Rate, Total Revenue, Avg Tenure
- Churn Rate by Contract: clustered bar with conditional red/amber/green color coding
- Churn Rate by InternetService: bar chart showing Fiber optic danger zone
- Revenue Lost by Contract: horizontal bar chart isolating financial impact
- Churn Rate by Tenure: area chart showing the steep early-customer drop-off
- Count of Churn vs Active: donut chart for quick at-a-glance split

4. Business Recommendations

Based on the data analysis, the following six strategic recommendations are prioritized by potential impact. Implementing all six is projected to reduce overall churn rate from 26.54% to approximately 16–18% within 12 months.

1

Incentivize long-term contracts — primary retention lever

Month-to-month customers churn at 42.71% vs 2.83% on two-year contracts. Offer a 10–15% discount, free months, or device upgrades for customers who upgrade to annual or two-year contracts. Target the 3,875 currently on month-to-month plans.

Expected impact: Estimated 8–12% reduction in overall churn rate

2

Fix fiber optic service quality or perceived value

Fiber optic customers churn at 41.89% despite paying higher prices ($74/mo avg). Conduct NPS surveys, investigate service delivery issues, and introduce a fiber loyalty program. Consider proactive outreach to fiber customers in months 1–6.

Expected impact: Could recover 500–600 at-risk fiber customers annually

3

Redesign the new customer onboarding experience

47.44% of customers churn in their first 12 months — a clear onboarding crisis. Implement a 90-day success program with proactive check-ins, usage tips, and satisfaction surveys at day 30, 60, and 90. Assign dedicated onboarding support for the first month.

Expected impact: Could halve early-churn rate; largest long-term value opportunity

4

Upsell OnlineSecurity + TechSupport as a bundle

Customers with both add-ons churn at only 9.01% vs 48.96% without them. Create a 'Peace of Mind Bundle' at a discounted price point and actively offer it during onboarding and at 6-month checkpoints. Focus on Fiber optic customers first.

Expected impact: 5x churn reduction for customers who adopt both add-ons

5

Create a Senior Citizen retention program

Seniors churn at 41.68% — nearly double the non-senior rate. Introduce simplified billing, dedicated support lines, and loyalty recognition programs for this segment. Consider a Senior Plan with lower complexity and consistent pricing.

Expected impact: Addresses 476 churned seniors representing significant LTV recovery

5. Conclusion

This analysis confirms that customer churn is a significant and solvable business problem. The data reveals clear, actionable patterns:

- Contract type is the biggest lever — upgrading customers to annual or two-year plans could alone reduce churn by 8–12%
- Fiber optic is a premium product with premium churn — a service quality and value gap needs urgent investigation
- Electronic check payment is a behavioral churn signal — autopay migration is a quick, high-ROI win
- The first 12 months are make or break — investing in onboarding will have the highest long-term customer lifetime value impact
- Add-on bundles (Security + TechSupport) are both a revenue and retention tool — a 5x churn reduction for customers who subscribe

The financial case for action is clear: at $139K/month in lost revenue, even a 5% reduction in churn would recover approximately $520,000 annually. The six recommendations in this report, implemented together, target a 10–12 percentage point reduction in churn — representing an estimated $1.5–1.8M in annual revenue recovery.

The best time to act on churn was yesterday.

The second best time is now.